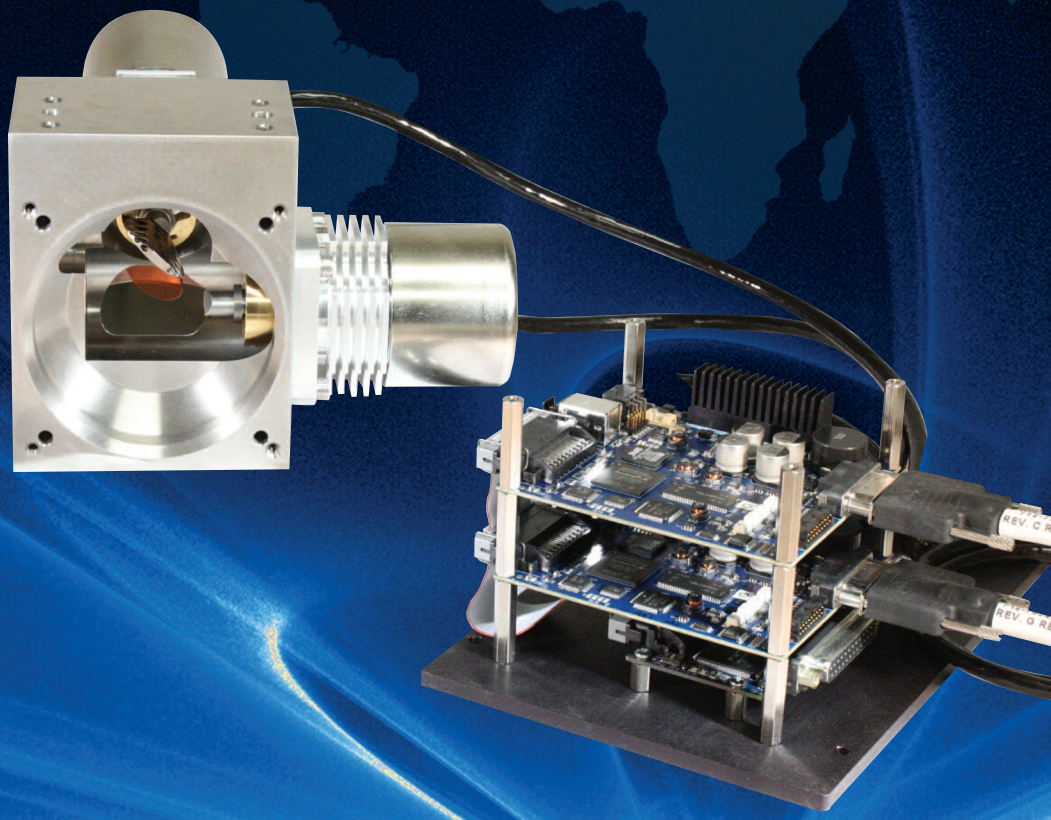
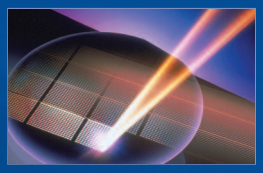
Cambridge Technology

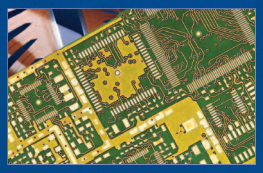
MOVING LIGHT, YEARS AHEAD.™

Lightning II™
Digital Scanning Platform



- TECHNOLOGY
- PERFORMANCE
- QUALITY
- VALUE
- RANGE OF PRODUCTS
- APPLICATIONS EXPERTISE

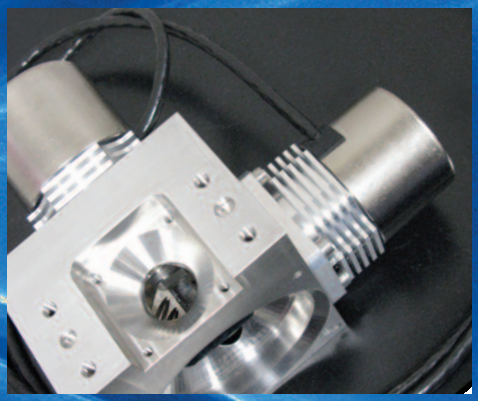




www.cambridgetechnology.com

CAMBRIDGE TECHNOLOGY

Moving Light, Years Ahead.



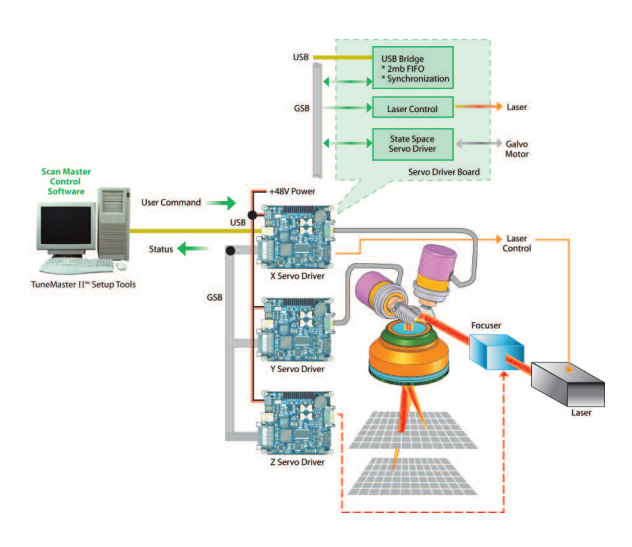
Lightning™ Digital Scanners

The Lightning II Digital Scanning System for Ultra High Accuracy and Speed Laser Material Processing Applications

Lightning II is an all digital integrated scanning system that has been designed for the highest levels of performance. This has been achieved by optimizing state-of-the-art technologies in laser / scanner control, communications, servos, drives, galvos, encoders and mirrors into one integrated platform. The System uses an integrated controller to achieve advanced laser / scanner synchronization control. Optionally, XY2-100 command input can be used from a traditional scan controller.

Lightning II Architecture. Six new elements that redefine state-of-the-art in optical scanning systems

1. LII Scan Control System – For the ultimate in synchronous modulated laser control
2. GSBus™ – 24-bit, 8-channel, high bandwidth bi-directional communications
3. LII State Space Servo – Anticipates laser position, faster than feedback
4. LII PWM Drive – Generates power instead of heat; 90% efficient amplifier
5. LII Encoder Galvos – Custom MicroE 24-bit encoder technology in a state-of-the-art galvo for breakthrough accuracy
6. Beryllium Mirrors – Shaped, lightened and balanced mirrors provide high speed and fast settling performance



A Typical Lightning II Digital Scanner System Includes:

- Two high performance MPM-20D (air cooled or water cooled) 24 bit encoder galvanometers
- High performance shaped Beryllium X,Y mirrors with Yag, CO_2 or other coating
- One precision engineered resonant structure X,Y motor block (single axis and 3-axis systems also available)
- Two Lightning II State Space digital servo drivers with PWM amplifiers and high bandwidth bi-directional 24 bit GSBus communications
- One XY2-100 to GSBus adapter interface (if needed for quick evaluation or traditional controller)
- LII Distributed Control System ™ (uses USB communications from PC to Lightning II GSBus for enhanced vector and laser control)
- LII set up & diagnostic toolset
- Interconnecting cables to adapter, servos and motors
- 1 meter long cooling hoses if needed (customer supplied water pump and coolant)
- Precision alignment and balancing of mirrors, motors and motor block
- Precision X,Y digital servo tuning to customer requirements
- Includes factory application support

WHY LIGHTNING II WITH SCANPACK?

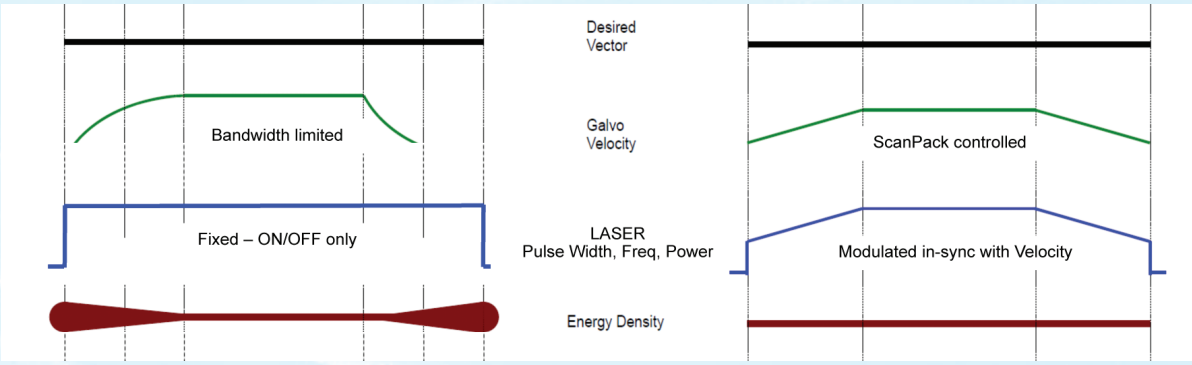
Traditional Control Scheme	Lightning II Control Scheme
• Mirror position is assumed-known based on position feedback; delays are required at the end of jumps (Jump Delay) and before changing mark direction (Stroke Delay) to ensure mirror position (= Stationary)	• Mirror position is always known based on complete modeling of scanner behavior; delays are NOT required at all
• Synchronization of laser firing to mirror motion is pre-programmed by the user; additional delays might be required (Laser-On Delay & Laser-Off Delay)	• Synchronization of laser control to mirror motion is dynamically calculated by the software during job execution
• The user combines process parameters (laser and scanner information) with pattern information (vectors & jumps) at the time of program creation, requiring re-programming for adaptation to varying process requirements (different laser / scanners / pattern)	• The software combines process parameters (laser and scanner information) with pattern information (vectors & jumps) dynamically during execution, allowing adaptation to varying process requirements (different laser / scanners / pattern) requires no re-programming
• One set of delays is defined based on worst-case requirements and is used for the entire job	• Motion is limited by maximum acceleration only, and is locally optimized by the software during job execution

HOW DO YOU MAINTAIN CONSTANT LASER POWER DENSITY WITH CHANGING VELOCITY?

- Lightning II with ScanPack maintains constant energy density by modulating the laser parameters in sync with the scanner's changing speed and acceleration

Traditional Control Scheme

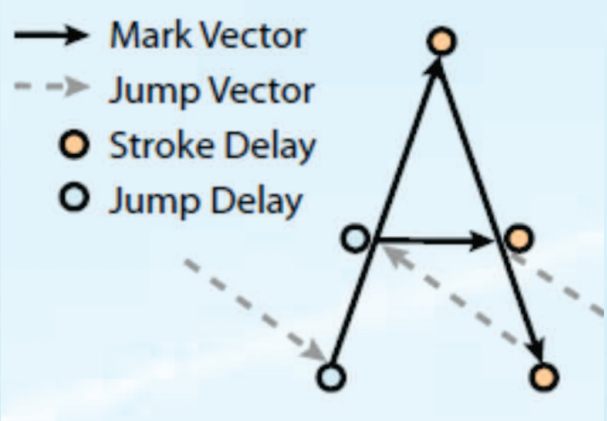
Lightning II with ScanPack Control



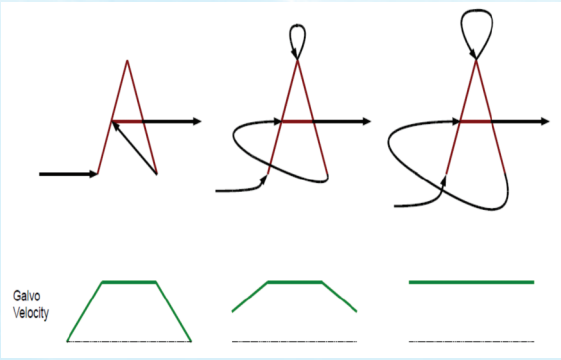
HOW DO YOU MAINTAIN SPEED AND ACCURACY AROUND SHARP CORNERS?

- Lightning II with ScanPack links vectors based on the scanner's acceleration limits and user-defined speed + accuracy constraints. 'Skywriting' automatically activates and adjusts extent of loop to optimize accuracy of joined vectors

Traditional Control Scheme



Lightning II with ScanPack Control



ScanPack generated vectors keep accuracy unchanged for different velocity profiles.

Lightning II Specifications

Galvo Motor	MPM-20D / 30D
Mirror Scan Angle	±20° Optical Typical Aperture Dependent
Repeatability	<2 µrad optical RMS
Dither	<2 µrad optical RMS
Thermal Drift	<2 µrad optical / °C
Long Term Drift	<10 µrad optical / 8hrs
Power Requirements	+15 to +48 VDC 3 Amps each continuous 6 Amps peak
Digital Communication	GSB / USB / XY2-100
Status Signals	Position Acknowledge System Ready
Operating Temp	15° to 35° C
Lightning II System Requirements	Windows XP, Windows 7

The Lightning II Digital Scanner System Models Available:

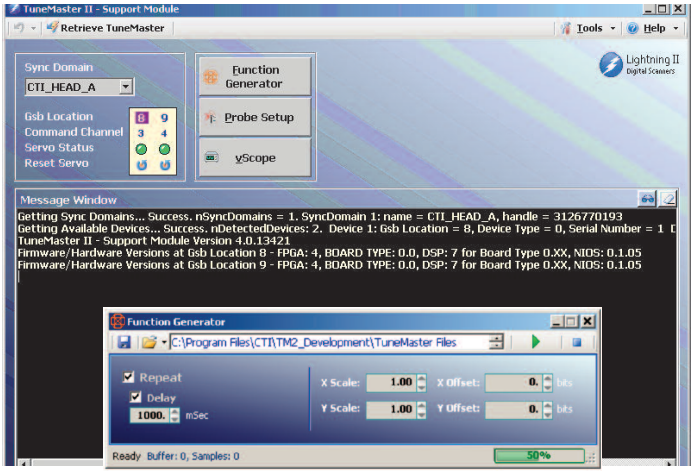
- Sizes available: 14, 20, 25, 27, 30 or 50mm beam aperture models
- Configurations: Systems are modular with mounting into your machine
- Wavelengths: Mirror coatings for CO_2, Yag, 532 and 355 lasers standard; custom available
- Single, 2- and 3-axis systems available

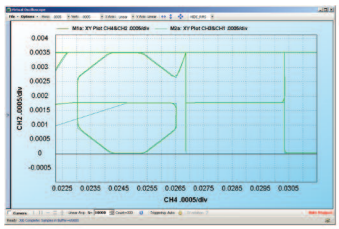
Lightning II is now offered with advanced SM1000 controller features allowing extended connectivity for system integrators:

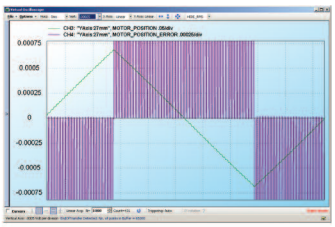
- 20-bit command resolution for higher accuracy
- True 2-dimensional motion tracking between scan-head and object for advance MOTF applications
- High level software interface (SM-API) to support customer system programming
- New software release to make Lighting II job creation easy
- Cutting-edge algorithms for process developers

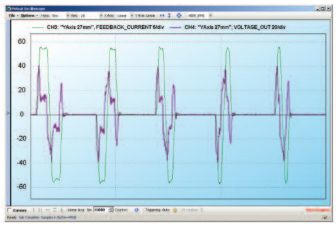
Lightning II Setup Tools Simplify Integration and Test

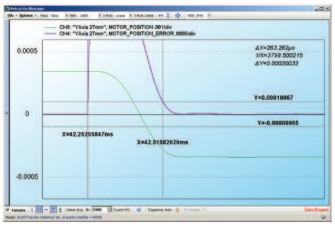
- Includes Function Generator and Virtual O'Scope to probe over 50 scanner+servo parameters over USB interface











Contact Cambridge Technology to learn more about our comprehensive line of products.

Components

2-Axis Scan Heads

Scan Control

3-Axis Scanning Systems

Lightning II Digital Scanning Platform

Cambridge Technology

MOVING LIGHT, YEARS AHEAD.™

125 Middlesex Turnpike • Bedford, MA 01730

P: (781) 266-5700

www.cambridgetechnology.com